

QuIP: Experimental Design for Expensive Simulators with Many Qualitative Factors via Integer Programming

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Joint work with Simon Mak

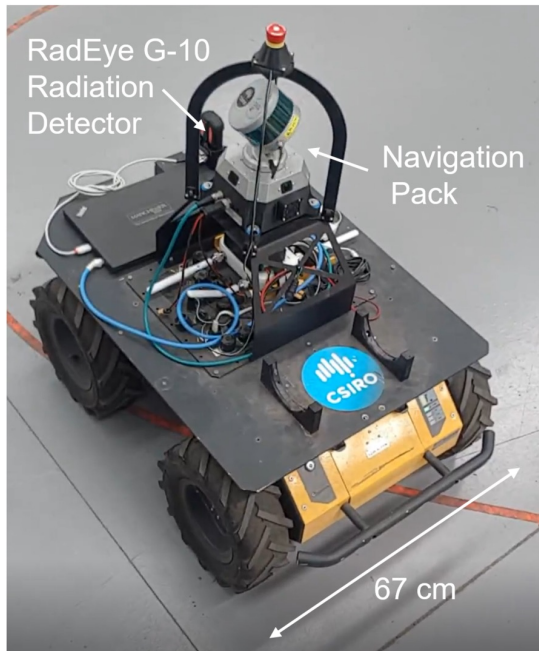
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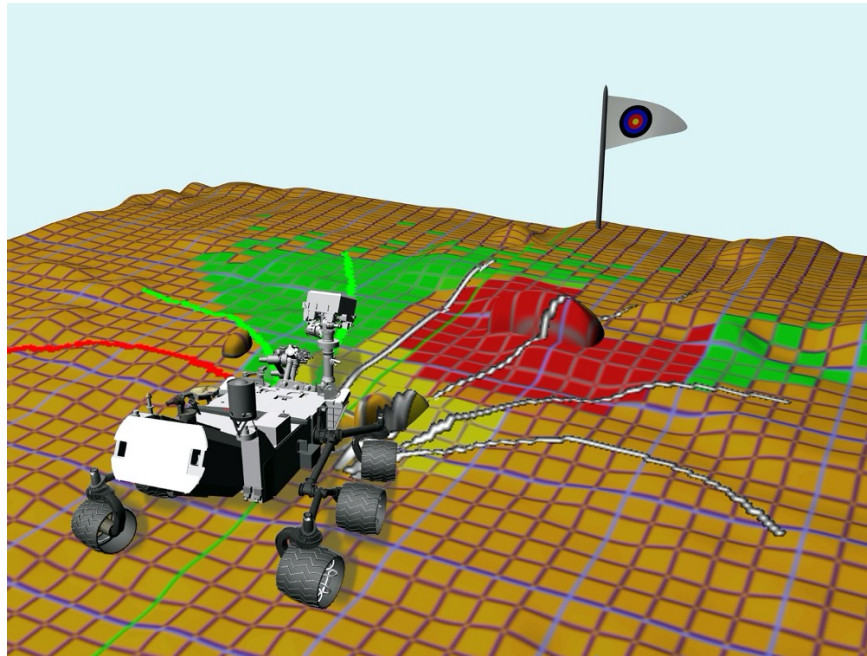
Motivation: Path planning problems

Goal: Find the **quickest navigation path** that takes the robot to destination

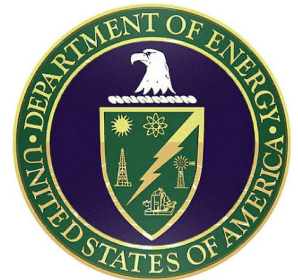
- **E.g., Path planning** of robotic vehicles for **radiation mapping**



Groves et al. (2021)

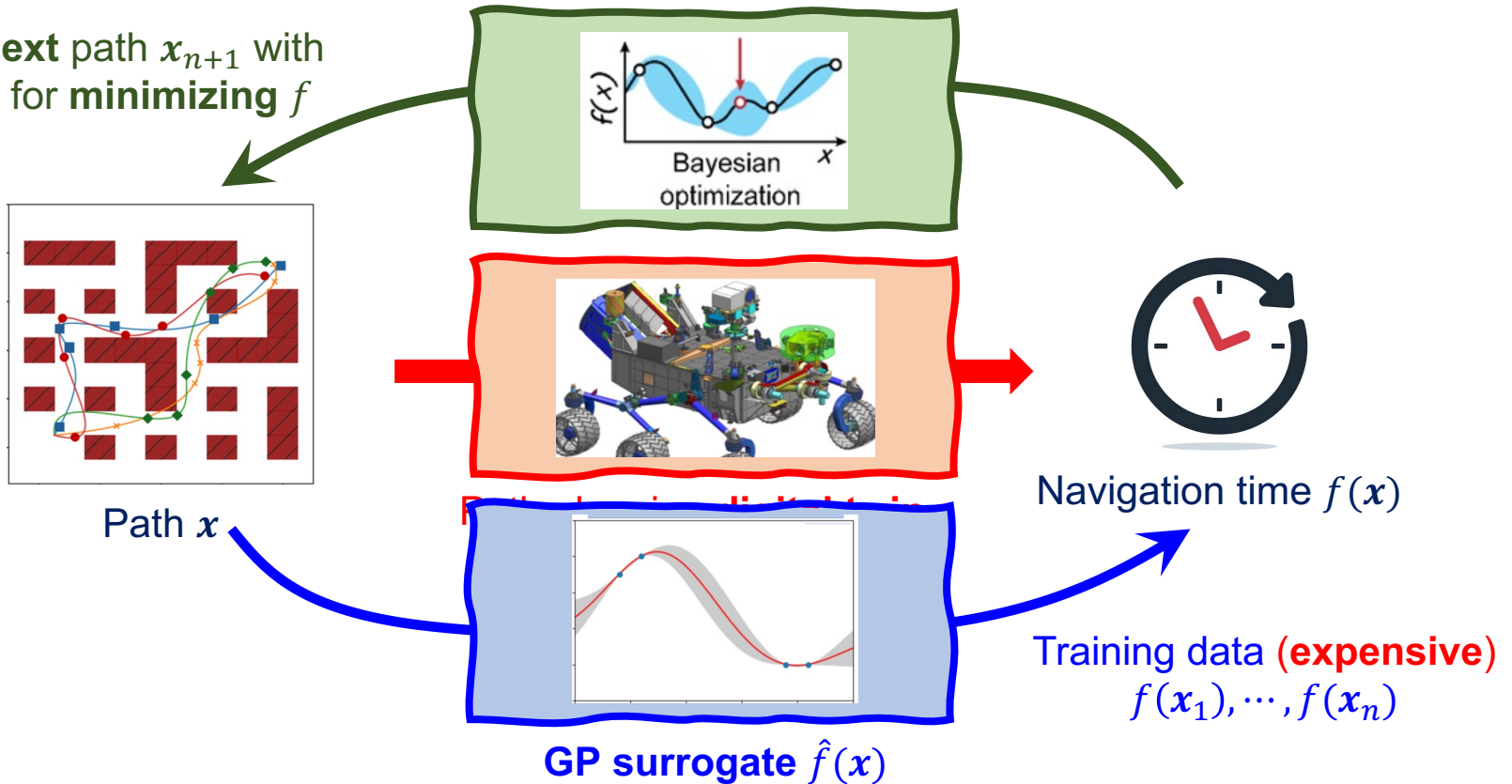


<https://www-robotics.jpl.nasa.gov/>



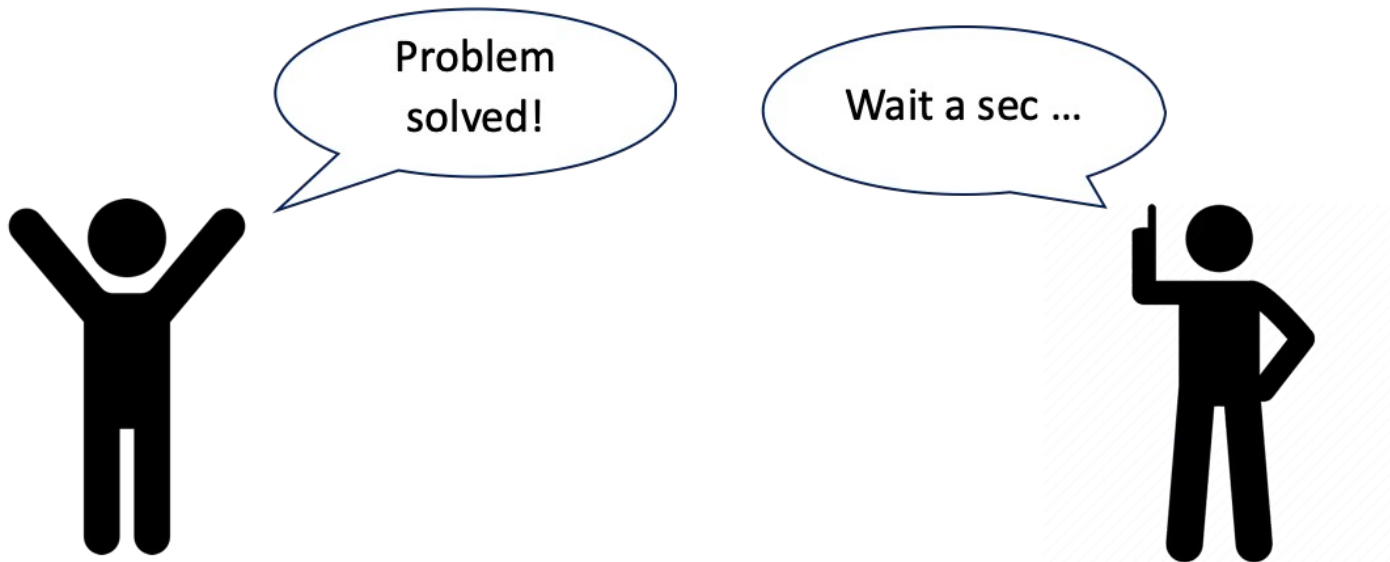
Experimental design

Identify **next** path x_{n+1} with potential for **minimizing** f



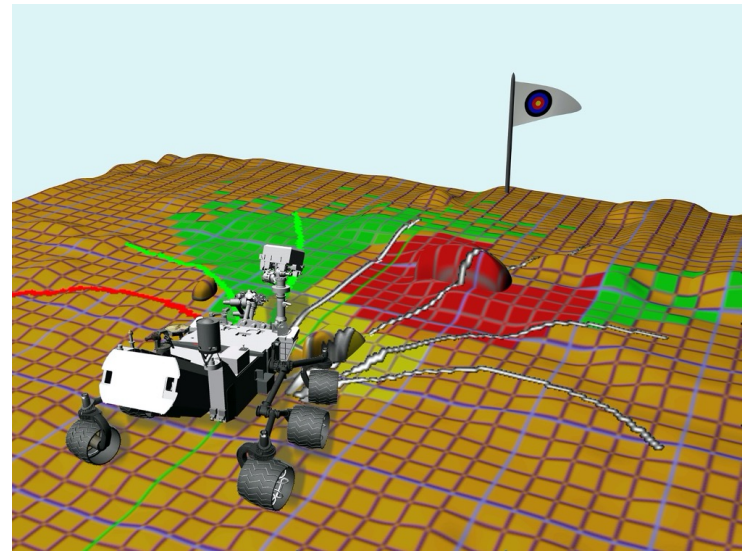
Traditional Approaches

- Dijkstra's algorithm and its variants (Dijkstra, 1976)
 - Particle swarm optimization (Kennedy and Eberhart, 1995)
 - Rapidly exploring random trees (LaValle and Kuffner, 2001)
- and so on



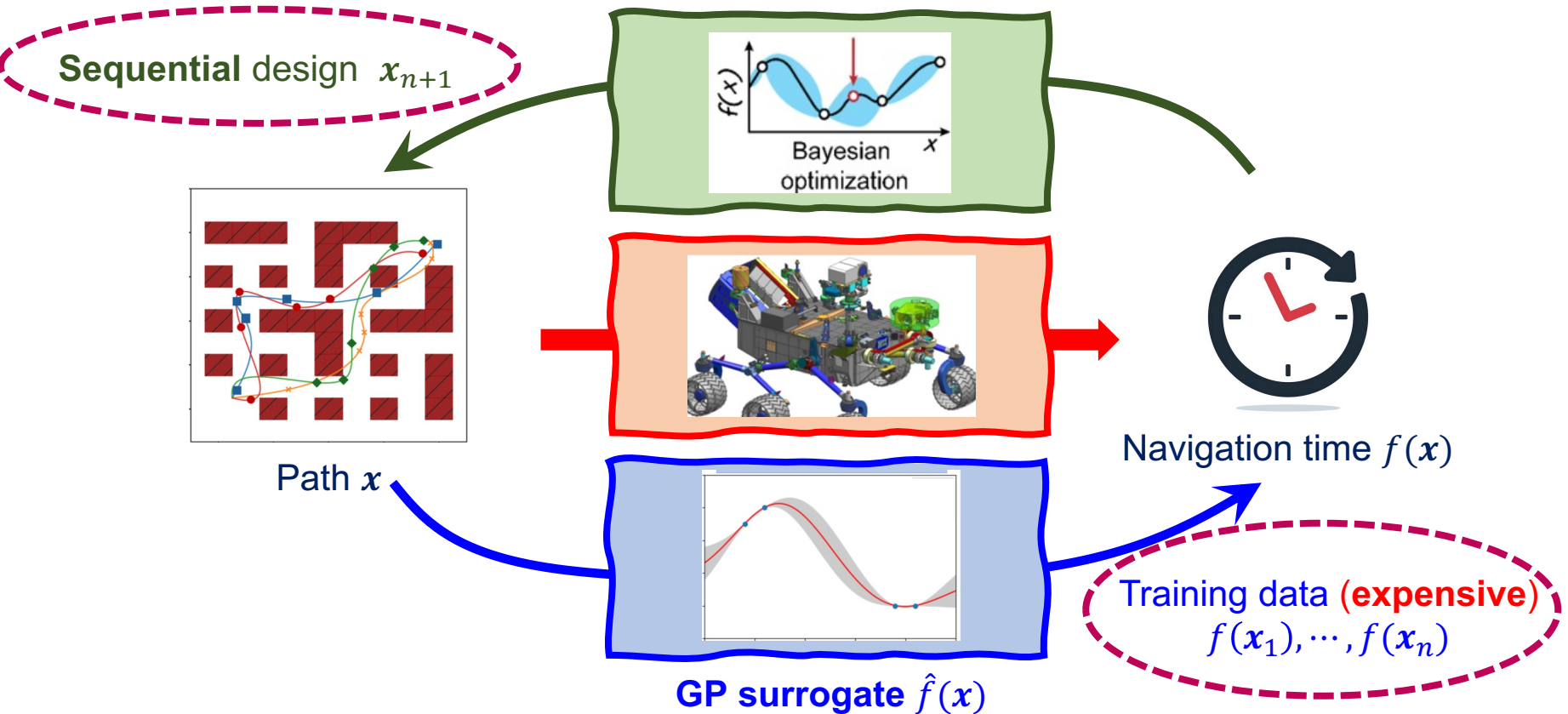
Challenges

- **Path planning** is now needed for **highly sophisticated** systems.
- Path planning using a **high-fidelity simulator/digital twin**:
 - **Costly multi-physics** simulations
 - $\mathcal{O}(\text{minutes})$ / path
 - Hinders **real-time** navigation
 - Traditional algorithms are not applicable.



<https://www-robotics.jpl.nasa.gov/>

Experimental design



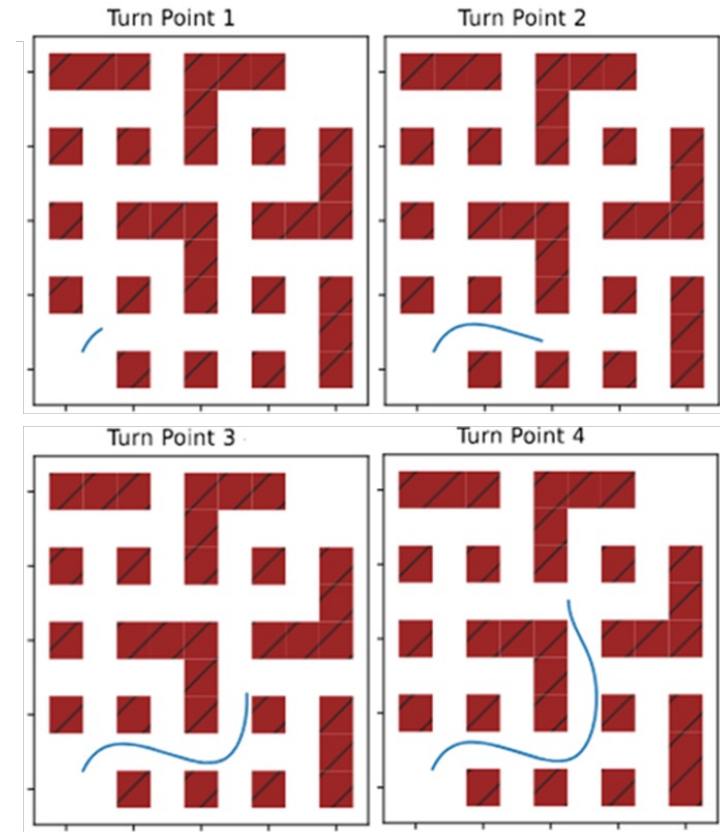
What about the **experimental design** problem?

- **Initial** experimental design
- **Sequential** experimental design

Discrete path planning

Parametrizing a navigation path $x = (x_1, \dots, x_d)$:

- $d \gg 1$ fixed **turn points**
- $M \gg 1$ discrete **actions** / turn point
 - E.g., turn angle, turn speed
 - Engineering & communication **constraints**
 - $x_l \in \{1, \dots, M\}$
- x consists of **many qualitative** factors:
 - M^d possible paths to investigate
 - **High-dimensional** experimental design



GPs for **qualitative** input factors:

- Qian et al. (2008): **exchangeable** kernel GP $f(\mathbf{x}) \sim \text{GP}(\mu, k(\cdot, \cdot))$ $\theta_l > 0$: **length-scale**

$$k(\mathbf{x}, \mathbf{x}') = \text{Cov}(f(\mathbf{x}), f(\mathbf{x}')) = \exp \left\{ - \sum_{l=1}^d \theta_l \mathbf{1}(x_l \neq x'_l) \right\}$$

- **Intuition**: $\text{Cov}(f(\mathbf{x}), f(\mathbf{x}')) \downarrow$ when paths $\mathbf{x}, \mathbf{x}'$ have many **dissimilar actions**
- **Simple** model, but **effective** for experimental design
- Zhang et al. (2020): **latent variable** GP (LVGP)
 - Qualitative factors mapped to **latent continuous** space

Initial design

- Question:**

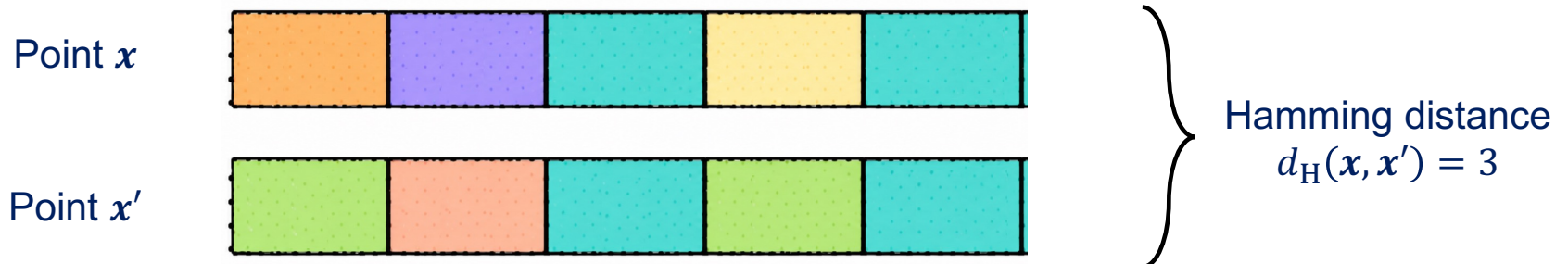
How to select **design points** $\mathcal{D} = \{x_1, \dots, x_n\}$ to maximize **learning** on f ?

- One **intuitive** design strategy:

- $d_H(x, x') = \sum_{l=1}^d \mathbf{1}(x_l \neq x'_l)$ is the **Hamming** distance
- Find the **Hamming-optimal** design $\mathcal{D}^*$ that **maximizes**:

$$\min_{x \neq x'} d_H(x, x')$$

- Intuition:** ensures **no** two points are **too close** (in Hamming distance)



Theorem: Assume $f \sim \text{GP}(\mu, k)$ follows the **exchangeable** kernel GP. Then the **Hamming-optimal** design $\mathcal{D}^*$ is **asymptotically D-optimal** for sampling f as the **length-scales** of the GP model $\theta \rightarrow \infty$. That is, $\mathcal{D}^*$ satisfies

$$\lim_{\theta \rightarrow \infty} \frac{1 - \det\{\Gamma_{\theta}(\mathcal{D}_{n,\theta}^*)\}}{1 - \det\{\Gamma_{\theta}(\mathcal{D}_n^*)\}} = 1,$$

where $\mathcal{D}_{n,\theta}^*$ is the D-optimal design under of the exchangeable GP with length-scale θ .

- This theory **bridges the gap** between **model-free designs** and **model-based optimal designs**!
 - **Model-free design**: robust to model-misspecification
 - **Model-based design**: optimal when model is correctly-specified
- Even better – the Hamming-optimal design can be formulated as an **assignment problem**!

Assignment problem (AP)

- Well-known **integer programming** problem with a rich history
- Aims to optimally **assign** workers to jobs such that total **utility** is **maximized**
- Many **generalizations**, e.g., **quadratic** assignment problem (QAP)

Formulation:

$$\max_I \sum_{i,i',j,j'} I_{ij} u_{ij}$$

$$\text{s. t. } \sum_i I_{ij} = 1 \text{ and } \sum_j I_{ij} = 1,$$

- Binary** decision variables $I_{ij} \in \{0,1\}$
 - 1 if worker i **assigned** to job j

	Job1	Job2	Job3	Job4
Worker 1	2	0	3	5
Worker 2	5	0	7	5
Worker 3	8	1	0	1
Worker 4	3	0	4	11

Assignment problem (AP)

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Advantages of AP:

1. Guaranteed convergence to **global optimum**
2. Providing **optimization gaps** (not available in heuristic algorithms)
3. Can be solved efficiently by state-of-art IP optimizers (e.g., Gurobi)

- We first convert a qualitative design to **dummy variables**.
- For example, consider an action set $= \{a, b, c, d\}$ and design $\mathbf{x} = (b, c, a, d, a)$. Then,

$$I(b, c, a, d, a) = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \end{matrix}.$$

- Each $I(\mathbf{x})$ must satisfy $\sum_{j=1}^m I(\mathbf{x})_{i,j} = 1$ for all i .
- The Hamming distance between designs can be represented as

$$d_H(\mathbf{x}, \mathbf{x}') = d - \text{tr}\{I(\mathbf{x})I(\mathbf{x}')^\top\}.$$

- Thus, we can represent a Hamming-optimal design problem as a **generalized AP problem**:

$$\begin{aligned} & \arg \max_{\mathcal{I}} \min_{i, i' \text{ with } i \neq i'} \{d - \text{tr}(\mathbf{I}_i \mathbf{I}_{i'}^\top)\} \\ \text{s.t. } & \sum_{k=1}^M I_{ijk} = 1, \quad \text{for all } i = 1, \dots, n \text{ and } j = 1, \dots, d, \\ & I_{ijk} \in \{0, 1\}, \quad \text{for all } i, j \text{ and } k, \end{aligned}$$

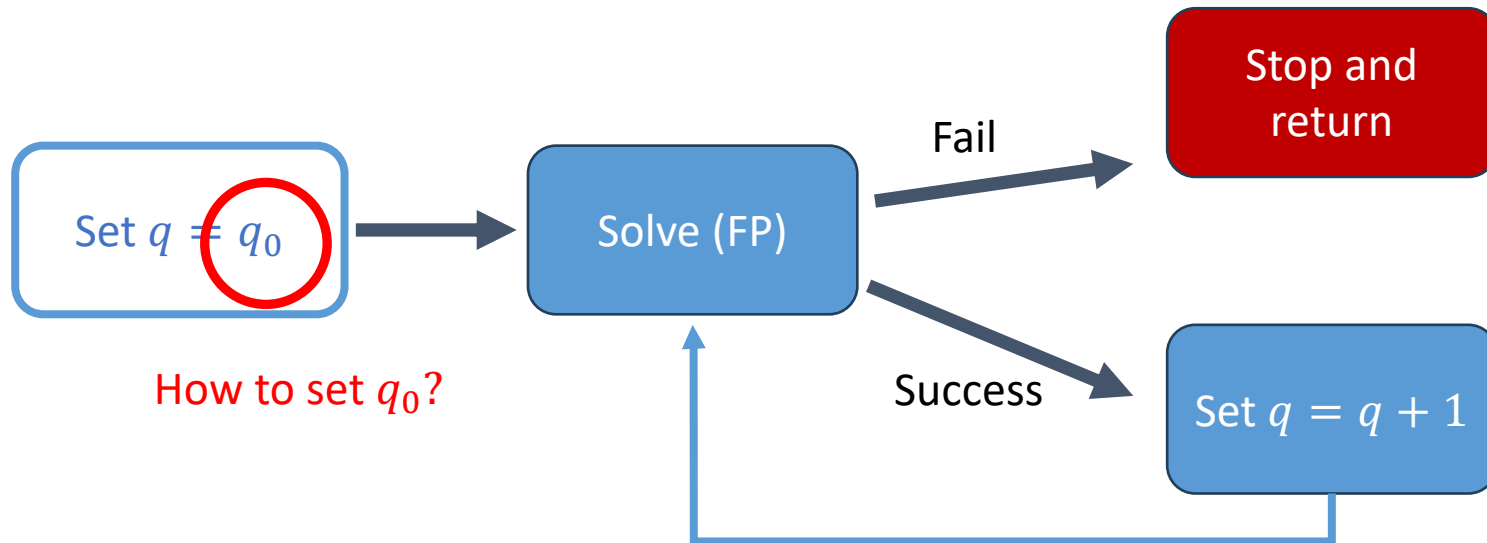
where $\mathbf{I}_i = [I_{ijk}]_{j=1}^d_{k=1}^M$ is the i –slice of the tensor $\mathcal{I}$ over its first dimension.

- It is a **non-convex optimization problem** involving $nd(M - 1)$ binary variables!

Optimization algorithm

We propose an **iterative feasibility program** (FP; Bertsimas and Tsitsiklis, 1997) to further speed up optimization time:

$$\begin{aligned} \arg \max_{\mathcal{I}} \quad & 1 \quad \text{s.t.} \quad d - \text{tr}(\mathbf{I}_i \mathbf{I}_{i'}^\top) \geq q, \quad \text{for all } i, i' = 1, \dots, n \text{ with } i \neq i', \\ & \sum_{k=1}^M I_{ijk} = 1, \quad \text{for all } i = 1, \dots, n \text{ and } j = 1, \dots, d, \\ & I_{ijk} \in \{0, 1\}, \quad \text{for all } i, j \text{ and } k. \end{aligned}$$



Lower bound for FP

Proposition: Consider the feasibility program (FP). Define the bound:

$$q_0(n, d, M) = \max \left\{ k \in \{1, \dots, d\} : \frac{M^d}{\sum_{l=0}^{k-1} \binom{d}{l} D_l} \geq n \right\}.$$

Then, for any choice of $q \leq q_0(n, d, M)$, the FP has at least one solution.

Proof: by the Gilbert-Varshamov bound (Gilbert, 1952; Varshamov, 1957)

Midtime takeaway

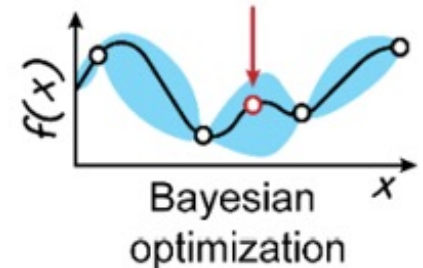
- Focus on **discrete path planning problems** with expensive simulators.
- Identified **two** experimental design problems:
 - **Initial design**
 - **Sequential design** for Bayesian optimization.
- Proposed **Hamming-optimal designs**
 - Proved that it is asy. D-optimal under exchangeable kernels.
- Formulated it into **assignment/integer programming problems**.
 - Guaranteed global optimization, provided optimization gaps, etc
- **Developed optimization algorithm** for solving the Hamming optimal design.

Sequential design

- **Goal:** Identify **next** point with potential for **minimizing** f (i.e., navigation time)
 - **Bayesian** optimization (Frazier et al. 2018, Chen et al. 2024 JASA)
- **Lower confidence bound** (LCB; Auer et al. 2002):

$$\mathbf{x}_{n+1} \leftarrow \arg \min_x \hat{f}(\mathbf{x}) - \lambda s(\mathbf{x})$$

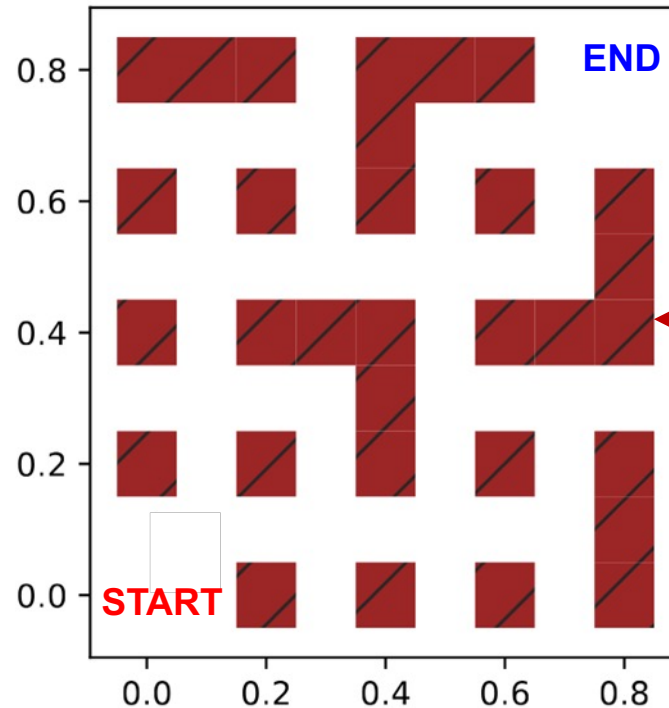
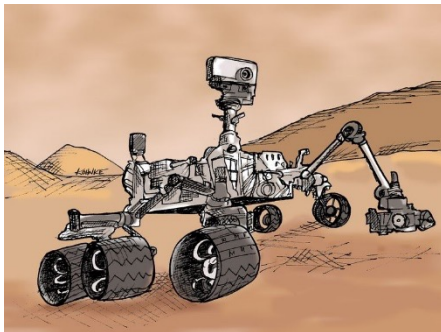
- **Exploitation:** find $\mathbf{x}$ with small **predicted** $\hat{f}(\mathbf{x})$
- **Exploration:** find $\mathbf{x}$ with high predictive **uncertainty** $s(\mathbf{x})$



Proposition: Assume $f \sim$ **exchangeable** kernel GP. Then the **LCB sequential design** can be formulated as a **generalized assignment problem**.

Case study: Path planning

Path planning case study:

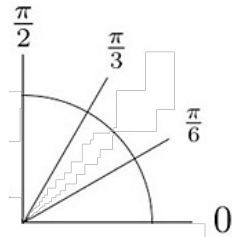


Swamps
(can pass through with
greatly reduced speed)

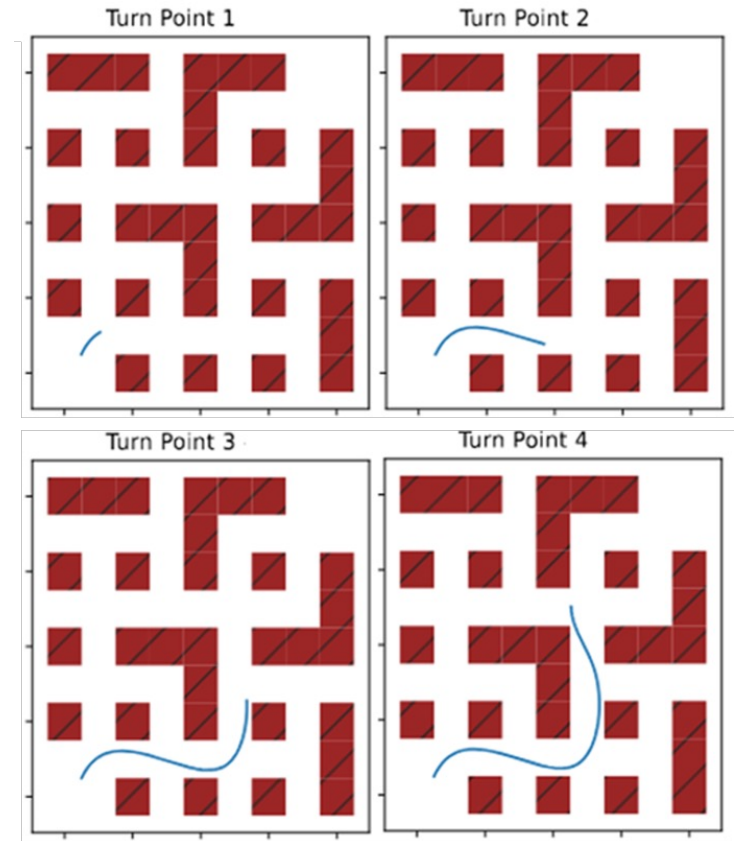
Case study: Path planning

Path planning case study:

- $d = 8$ path **turn points/decisions**
- $M = 9$ navigation **choices** at a turn point:
 - Two **speed** choices (low, high)
 - Four **angle** choices (from $0 - \frac{\pi}{2}$ rad.)



- ... or simply **stay put**
- $9^8 \approx 43$ mil. paths to **investigate!**



Case study: Path planning

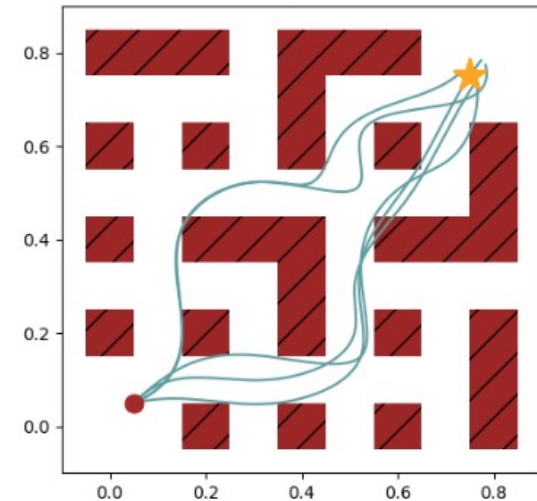
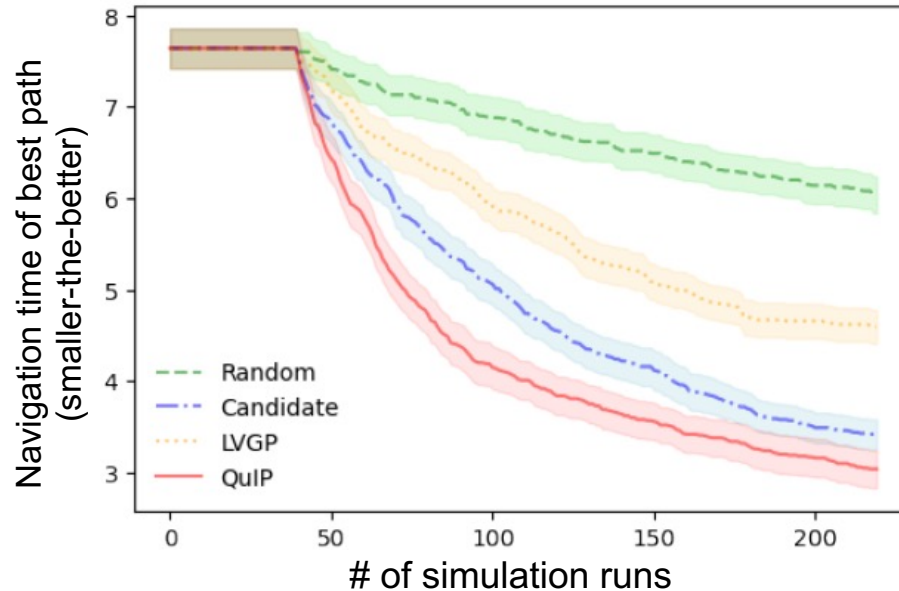
- **Simulator**: Python path planning digital twin (Wang et al. 2018)
- Proposed **QulP** approach (**exchangeable** kernel GP):
 - **Initial** design (Hamming-optimal): $n = 30$ points
 - **Sequential** design (LCB): $n_{\text{seq}} = 200$ points
 - Designs **globally** optimized via Gurobi

Benchmarks:

- **Random** design points
- **Candidate: Same** set-up as QulP, but designs optimized via random **candidate set**
- **LVGP**: Latent variable GP (Zhang et al. 2020)
 - More **complex** model, design optimization via random **candidate set**

Case study: Path planning

Path planning case study:



Examples of low cost paths discovered by QuIP

- **QuIP** performs **better** than **candidate** approach
- **LVGP** (using more **complex** model) performs **worse** than **QuIP**
 - Importance of a **well-optimized** design under limited budget

- **Many qualitative factors** create a challenging experimental design problem.
- **QuIP** turns experimental design **into integer programming problems** that
 - Enable global optimization
 - Provide optimality gaps
 - Can be solved by efficient IP solvers.
- We develop **explicit IP formulations** for both the initial and sequential design problems.
- **Good design optimization can matter as much as model complexity!**

Thank you!

Liu, Y. C., & Mak, S. (2026). QulP: Experimental design for expensive simulators with many Qualitative factors via Integer Programming. *Journal of Computational and Graphical Statistics*, to appear



- Bertsimas, D., & Tsitsiklis, J. N. (1997). *Introduction to linear optimization* (Vol. 6, pp. 479-530). Belmont, MA: Athena scientific.
- Gilbert, E. N. (1952). A comparison of signalling alphabets. *The Bell system technical journal*, 31(3), 504-522.
- Li, K., Mak, S., Paquet, J. F., & Bass, S. A. (2026). Additive multi-index Gaussian process modeling, with application to multi-physics surrogate modeling of the quark-gluon plasma. *Journal of the American Statistical Association*, 121(553), 44-59.
- Liyanage, D., Everett, D., Chattopadhyay, C., & Heinz, U. (2022). Prehydrodynamic evolution and its impact on quark-gluon plasma signatures. *Physical Review C*, 105(6), 064908.
- Ji, Y., Yuchi, H. S., Soeder, D., Paquet, J. F., Bass, S. A., Joseph, V. R., ... & Mak, S. (2024). Conglomerate multi-fidelity Gaussian process modeling, with application to heavy-ion collisions. *SIAM/ASA Journal on Uncertainty Quantification*, 12(2), 473-502.
- Mak, S., Sung, C. L., Wang, X., Yeh, S. T., Chang, Y. H., Joseph, V. R., ... & Wu, C. J. (2018). An efficient surrogate model for emulation and physics extraction of large eddy simulations. *Journal of the American Statistical Association*, 113(524), 1443-1456.
- Qian, P. Z. G., Wu, H., & Wu, C. J. (2008). Gaussian process models for computer experiments with qualitative and quantitative factors. *Technometrics*, 50(3), 383-396.
- Stokes, Z., Wong, W. K., & Xu, H. (2024). Metaheuristic solutions to order-of-addition design problems. *Journal of Computational and Graphical Statistics*, 33(3), 1006-1016.
- Varshamov, R. R. (1957). Estimate of the number of signals in error correcting codes. *Doklady Akad. Nauk, SSSR*, 117, 739-741.
- Wang, Z., Gehring, C., Kohli, P., & Jegelka, S. (2018, March). Batched large-scale Bayesian optimization in high-dimensional spaces. In *International conference on artificial intelligence and statistics* (pp. 745-754). PMLR.
- Zhang, Y., Tao, S., Chen, W., & Apley, D. W. (2020). A latent variable approach to Gaussian process modeling with qualitative and quantitative factors. *Technometrics*, 62(3), 291-302.

Simulation – Initial design

Setup:

- Initial design size $n = 50$
- Number of qualitative levels: $M = 5, 10, 20, 30$
- Number of qualitative variables: $d = M + 15, M + 25$

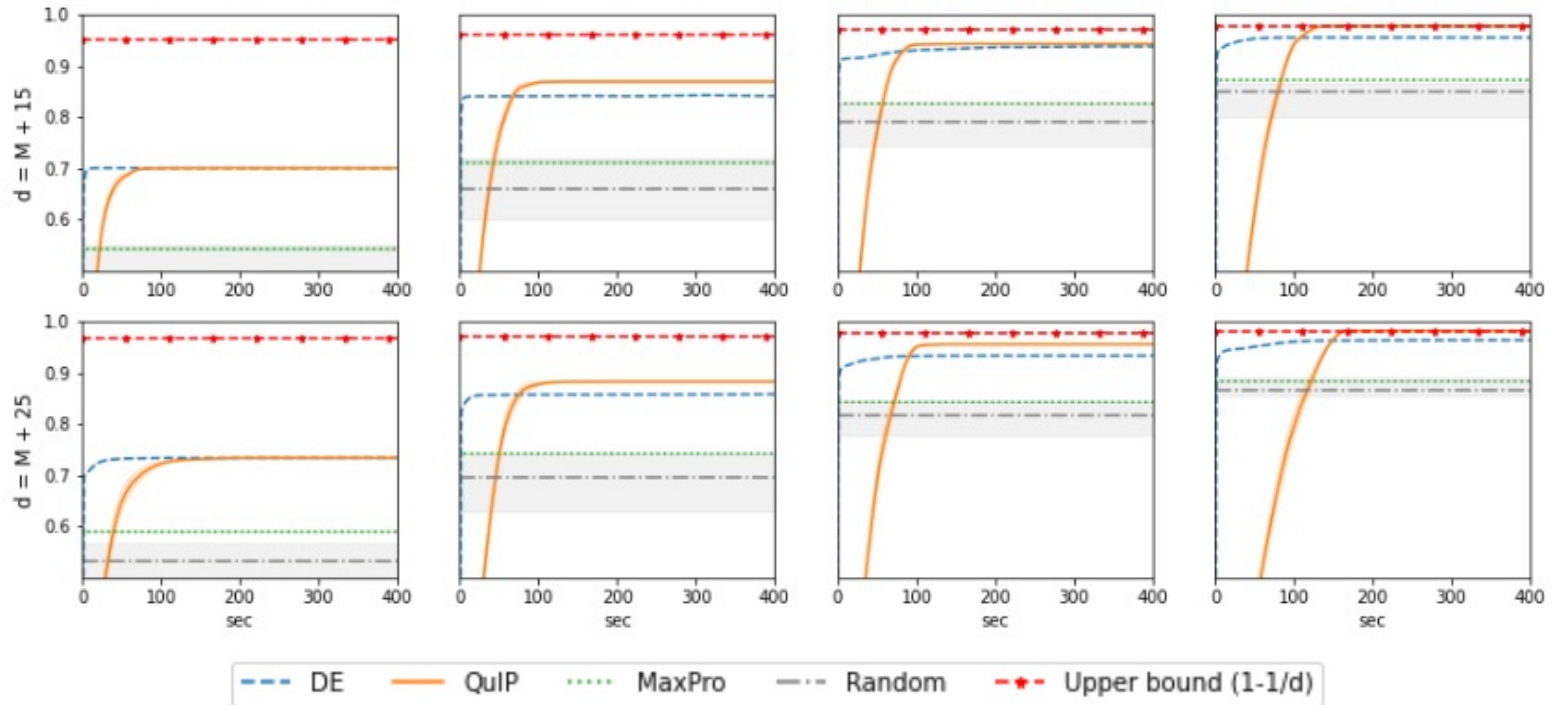
Benchmarks:

- **Random** design
- **Differential Evolution (DE)** – a metaheuristics algorithm (Stokes et al., 2023)
- **MaxPro** design (Joseph et al., 2020)
- **QuIP** - our method (Hamming-optimal design)

Metrics: minimum pairwise distance by optimization time (sec)

Simulation – Initial design

The larger (the minimum pairwise distance) is, the better.



- **Heuristics-based** algorithms tend to **stuck at local optimum**
- **QuIP (our method)** is able to achieve **better maximin designs** in almost all scenarios **with shorter time**.